

Chapter 3. **GROUPS**

3.1 What is a group?



Today we will

- Learn what a **GROUP** is, and
- See a few examples.

What is a group?

A **group** $(G, \cdot)$ is a set G equipped with a binary operation¹ $\cdot$ satisfying the following axioms:

$$\cdot : G \times G \rightarrow G$$

(G1) G is closed under $\cdot$. $a \cdot b \in G$ for all $a, b \in G$.

(G2) The **associative law** holds: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for every $a, b, c \in G$.

(G3) There is a unity 1 for G . Then $1 \cdot a = a \cdot 1 = a$ for all $a \in G$.

(G4) Every element of G has an inverse in G . In other words, for every $a \in G$ there is an element $b \in G$ such that $a \cdot b = b \cdot a = 1$

The group G is called **abelian** if, in addition, it satisfies

(G5) The operation $\cdot$ is commutative: $a \cdot b = b \cdot a$ holds for every $a, b \in G$.

If G is finite, the number $|G|$ is called the **order** of G .

Some important facts!

- When proving results, we will mostly write groups using a **MULTIPLICATIVE OPERATION**.
- Many well-known groups are additive.
- For **ADDITIVE GROUPS**:
 - The **unity element** is denoted as **0**; we call it the **ZERO**.
 - The **inverse of an element x** is denoted **-x**, called the **NEGATIVE of x**.

Examples 3.1.

trivial group

1. $\{1\}$, $\{1, -1\}$, $\{1, -1, i, -i\}$ are all abelian groups of (complex) numbers under multiplication. Notice that -1 is self-inverse, while i and $-i$ are inverses of each other.
2. $\mathbb{Q} \setminus \{0\}$, $\mathbb{R} \setminus \{0\}$, $\mathbb{C} \setminus \{0\}$ are all abelian groups under multiplication. In each case the inverse of an element a is $a^{-1} = 1/a$.
3. The sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are all abelian groups under addition. In each case the identity element is 0, and the inverse of a is its opposite $-a$.
4. Let $n \geq 2$ be an integer. Then $\mathbb{Z}_n$ is an additive abelian group with zero $\bar{0}$ and the negative of $\bar{a}$ being $-\bar{a} = \overline{-a}$.

$$\mathbb{Z}_n = \{\bar{a} \mid a \in \mathbb{Z}\} \stackrel{+}{=} \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$$

In Theorem 1.42 we saw that the modular addition has the following properties:

$$\bar{a} + \bar{b} = \overline{a+b}$$

$$(G2) \quad \bar{a} + (\bar{b} + \bar{c}) = (\bar{a} + \bar{b}) + \bar{c}$$

$$(G3) \quad \bar{a} + \bar{0} = \bar{a}$$

$$(G4) \quad \bar{a} + \overline{-a} = \bar{0}$$

$$(G5) \quad \bar{a} + \bar{b} = \bar{b} + \bar{a}$$

Notice that $\mathbb{Z}_n$ is closed under the modular addition by Theorem 1.34 (2).

(*) If $a \in \mathbb{Z}$, then $[a]_n = [r]_n$ for some $0 \leq r \leq n-1$.

5. Let R denote $\mathbb{Z}_m, \mathbb{Z}, \mathbb{Q}, \mathbb{R},$ or $\mathbb{C}$. Then the set $U(M_n(R), \cdot)$ of invertible $n \times n$ matrices over R , that is, the matrices $A \in M_n(R)$ such that $\det(A)$ is a unit in R , is a group called the **general linear group of degree n over R** , denoted $GL_n(R)$. Moreover:

It has a
multiplicative
inverse

If $R = \mathbb{Q}, \mathbb{R},$ or $\mathbb{C}$ then $GL_n(R) = \{A \in M_n(R) \mid \det(A) \neq 0\}$

$GL_n(\mathbb{Z}) = \{A \in M_n(\mathbb{Z}) \mid \det(A) = \pm 1\}$

$GL_n(\mathbb{Z}_m) = \{A \in M_n(\mathbb{Z}_m) \mid \det(A) = \bar{a}, \text{ where } \gcd(a, m) = 1\}$

When we refer to the groups $\mathbb{Z}_n, \mathbb{Z}, \mathbb{Q}, \mathbb{R},$ or $\mathbb{C}$, we mean the additive group.

$$\det(AB) = \det(A)\det(B)$$

$$\det(A^{-1}) = \frac{1}{\det(A)} = \det(A)^{-1}$$

6. The set S_n of all permutations of $\{1, 2, \dots, n\}$ is a group under composition, called the **symmetric group of degree n** .

An alternating group of degree n .

Definition 3.3. Let $G_1, G_2, \dots, G_n$ be groups. Their Cartesian product $G_1 \times G_2 \times \dots \times G_n$ with the **component-wise operation** defined by

$$(g_1, g_2, \dots, g_n) \cdot (g'_1, g'_2, \dots, g'_n) = (\overset{\uparrow}{g_1 g'_1}, \overset{\uparrow}{g_2 g'_2}, \dots, g_n g'_n),$$

where $g_i g'_i$ refers to the product in the group G_i for each i , is a group called the **direct group** of the G_i . In fact, one can easily check that the unity of $G_1 \times G_2 \times \dots \times G_n$ is $(1, 1, \dots, 1)$ and the inverse of the element $(g_1, g_2, \dots, g_n)$ is $(g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$.

EXAMPLE: Take $G = \mathbb{Z}_2$, and let us take a look to the direct product of two copies of $\mathbb{Z}_2$:

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1}), (\bar{1}, \bar{0}), (\bar{1}, \bar{1})\}$$

$$\mathbb{Z}_2 = \{\bar{0}, \bar{1}\}$$

Operation in $\mathbb{Z}_2 \times \mathbb{Z}_2$:

$$(\bar{a}, \bar{b}) + (\bar{c}, \bar{d}) = (\bar{a} + \bar{c}, \bar{b} + \bar{d}) = (\overline{a+c}, \overline{b+d})$$

Identity element $(\bar{0}, \bar{0})$

Opposite of $(\bar{0}, \bar{1})$ is $(\bar{0}, \bar{1})$.